

Self-attention Handwriting Generative Model

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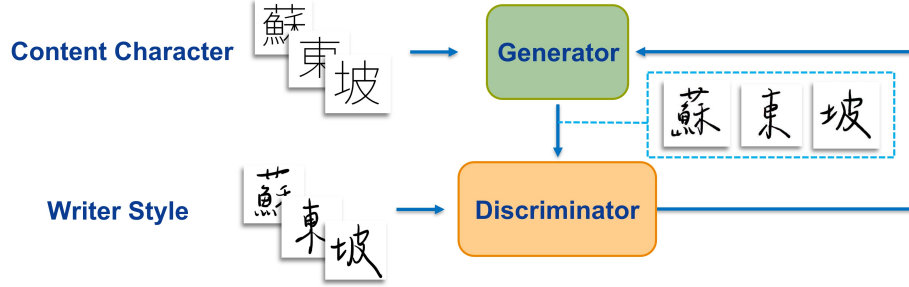


Figure 1: The architecture of zi2zi-self-attention uses a generator and a discriminator to generate handwritten style fonts.

Abstract

Recent advancements in font generation models include GANs, diffusion models, and transformers. This study introduces a GAN-based model, zi2zi self-attention, which enhances the zi2zi model by incorporating Residual Blocks (ResNet Block) and Self-Attention Layers in the encoder and decoder. These enhancements improve the ability to capture complex font details, mimicking the writer's style while preserving the source image's content. The study also introduces Perceptual Loss, using high-level features from the VGG-19 network to improve visual consistency and quality. The Adam optimizer with appropriate learning rates and Beta parameters is employed to enhance convergence speed and stability. For comparison, four other mainstream models are evaluated: MX-Font, CF-Font, SDT, and Font-Diffuser. These models differ from zi2zi in their focus on extracting content characters and writing style features rather than solely using adversarial networks. Experimental results show the potential and effectiveness of zi2zi-Self-Attention in generating high-quality, visually appealing handwritten fonts.

Keywords

Handwriting, Font Generation, GAN, Self-attention

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1 Introduction

Handwritten font generation methods are a type of generative AI focused on separating content and style information to achieve flexible and refined generation effects. Generative adversarial networks (GANs) generate more realistic results through the adversarial interaction between generators and discriminators. Diffusion models learn and separate real images by adding noise and a reverse diffusion process. Transformers utilize multi-layer feature extraction techniques to separate and recombine content and style information, thereby generating high-quality, diverse font images from a small number of samples. However, these methods often fall short in detail processing, particularly in reproducing complex strokes and subtle stylistic features, and they require large and diverse training datasets along with time-consuming data augmentation techniques.

Few-shot font generation aims to generate a large number of unwritten characters from a few samples, accurately capturing the writer's style features. Based on this, the motivation of this study is to improve the zi2zi conditional adversarial network font generation model. Unlike mainstream content-style separation methods, zi2zi focuses on style transfer of individual characters through the adversarial interaction between generators and discriminators, with a relatively simple and user-friendly architecture that reduces reliance on large datasets. This study has made several improvements to the zi2zi model, including the addition of residual blocks, self-attention mechanisms, and perceptual loss, naming the improved model zi2zi self-Attention, to better capture the detailed features of handwritten fonts and enhance generation quality. The study compares five existing font generation models, including zi2zi [He 2018], MX-Font [Park et al. 2021], CF-Font [Wang et al. 2023], SDT [Dai et al. 2023], and Font-Diffuser [Yang et al. 2024]. Additionally, this study develops a font packaging process, collecting and digitizing handwritten font files from 54 writers, thereby advancing the development of font generation technology.

2 Method

zi2zi is a conditional adversarial network transformation model. In this study, we improved it and named the enhanced model zi2zi Self-Attention to improve the effectiveness and quality of font generation. As shown in Figure 2, the red boxes highlight the improvements, which include the addition of residual blocks [He et al. 2016] and self-attention mechanisms [Vaswani et al. 2017] in the encoder and decoder. The residual blocks introduce shortcut connections that help better transmit gradients in deep neural networks, effectively addressing the gradient vanishing problem in deep networks while preserving essential features of characters. The self-attention mechanism captures long-range dependencies, allowing the network to better understand global information in font images. This enhances the coherence and consistency of generated fonts, especially when dealing with complex glyph structures, improving the generation results. We used the Adam optimizer during training. Additionally, to better measure the quality of generated fonts, we incorporated perceptual loss [Johnson et al. 2016], which extracts features using the VGG-19 network for comparison and evaluation. This helps the model better maintain the details and structure of generated fonts, making the results visually closer to real fonts. The figure 3 shows the comparison results of zi2zi self-attention with other methods, while Table 1 presents the evaluation results using assessment tools, demonstrating the effectiveness of this method.

Moreover, we developed a font packaging process that combines the characters generated by the model with the original input style character images into a font file. This process involves converting bitmap images to vector graphics, and then processing the fill-rule of the vector graphics. By doing so, we can convert PNG images to TTF fonts. This integration allows for seamless usage of generated fonts in various applications, enhancing their practical utility.

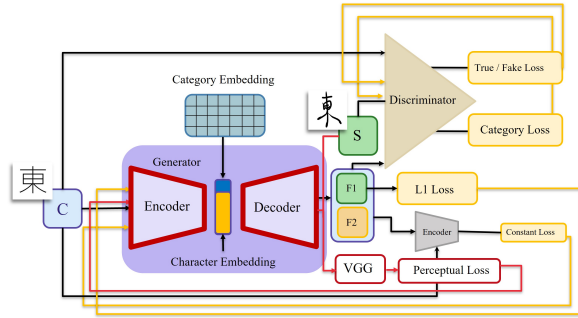


Figure 2: The architecture of zi2zi self-attention.

3 Conclusion

This study evaluates various font generation models to improve the quality and efficiency of handwritten font generation. Firstly, we enhanced the existing zi2zi handwritten font generation method by adding residual blocks and self-attention mechanisms, improving the font quality and accurately capturing handwriting styles. The results indicate that our method performs well across multiple metrics. However, due to the emphasis on preserving font details and structure, the model occasionally produces noise or extraneous

zi2zi	撞像碳嫖澳僻瞧
MX-Font	撞像碳嫖澳僻瞧
SDT	撞像碳嫖澳僻瞧
CF-Font	撞像碳嫖澳僻瞧
Font Diffuser	撞像碳嫖澳僻瞧
zi2zi-Self-Attention	撞像碳嫖澳僻瞧
Ground Truth	撞像碳嫖澳僻瞧

Figure 3: Comparison of our method and other models.

Table 1: Evaluations of generated outputs by different models

Methods	L1 Loss ↓	RMSE ↓	LPIPS ↓	SSIM ↑	FID (%) ↓
zi2zi	12.6	31.94	32.09	69.54	31.47
MX-Font	11.44	29.71	31.62	71.75	46.50
CF-Font	6.67	23.00	15.10	67.08	29.35
SDT	19.08	41.42	49.71	56.79	254.37
Font-Diffuser	14.58	33.79	37.72	64.86	69.29
zi2zi-Self-Attention	11.38	31.07	30.16	71.88	28.39

strokes when generating simple characters. As a result, CF-Font outperforms our method L1 Loss, RMSE and LPIPS. Additionally, we systematically collected and processed a dataset of 4,808 Chinese characters handwritten by 54 participants, creating a diverse and high-quality handwriting dataset. This dataset is crucial for improving the training effectiveness of generation models and the final font quality. The process also combined the generated characters into a complete font file, making the font generation process more comprehensive. These efforts provide valuable references for future font generation research and offer technical support and data foundation for high-quality font generation in practical applications.

Acknowledgments

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